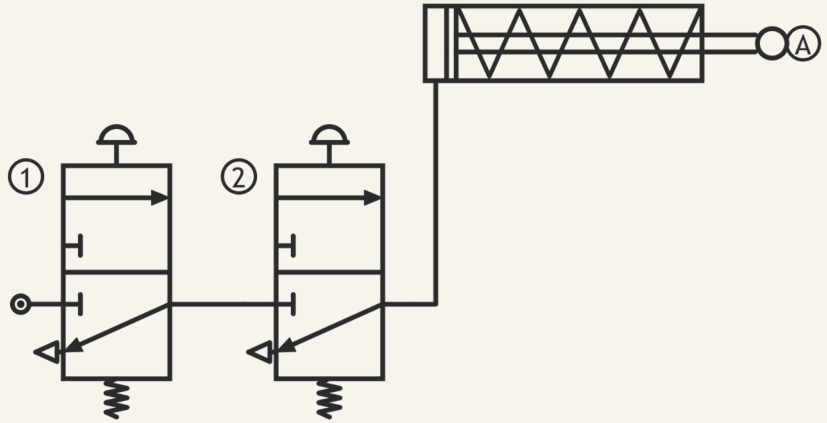
AND and OR Control

Both hands on the buttons. Why can't the press hurt them?



National 5 Engineering Science

2017 N5 Engineering Science: scan the code and try these.

Q11(b)

Page 15. Sketch it in your jotter.

Scan for the paper

Marking scheme

2023 N5 Engineering Science: scan the code and try these.

Q3

Page 4. Answer in your jotter.

Scan for the paper

Marking scheme

By the end of this lesson… (p.19)

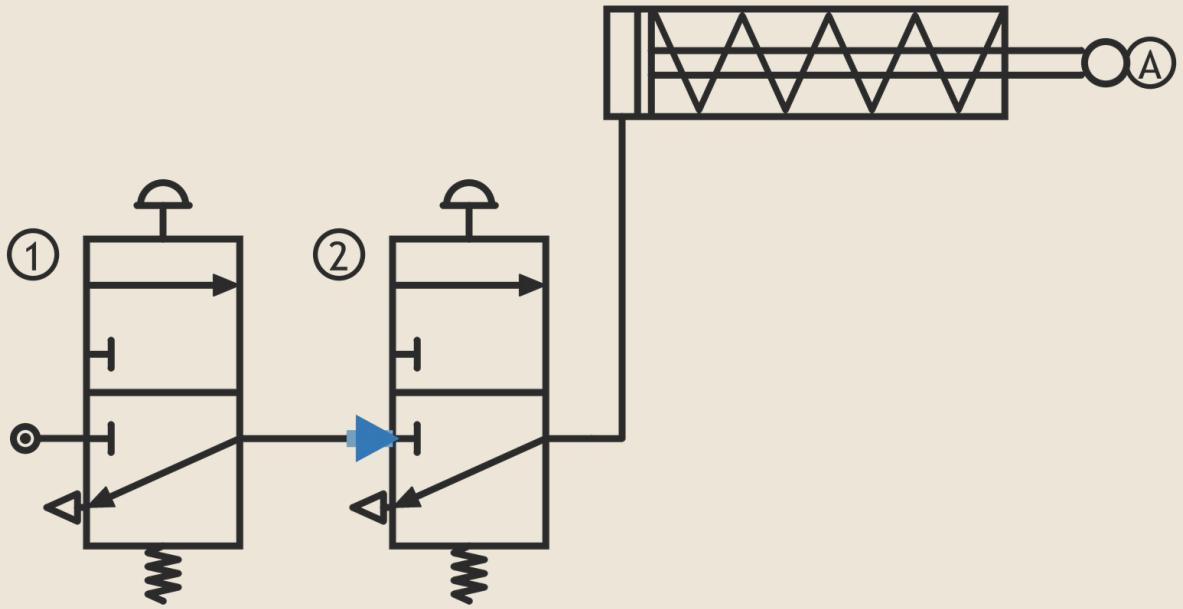


1. I can **pipe** two valves in series to give AND control.
2. I can **pipe** two valves and a shuttle valve to give OR control.
3. I can **state** the type of control a circuit uses, from its piping.
4. I can **explain** why two-hand AND control keeps an operator safe.

AND Control (p.19)

AND control: the piston moves only when valve 1 **and** valve 2 are both actuated.

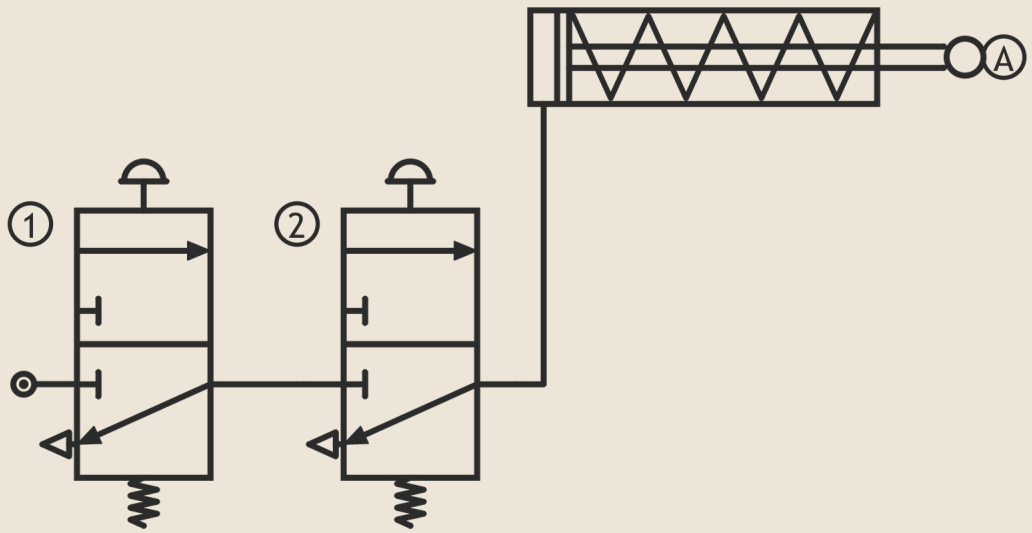
The valves are piped in **series**: the output of valve 1 is the main air supply for valve 2.



Two-Hand Control (p.19)

Two-hand control is AND control. Both hands must be on the buttons, so they are kept **away** from the piston.

The two buttons are **far apart**, each with a **guard**.

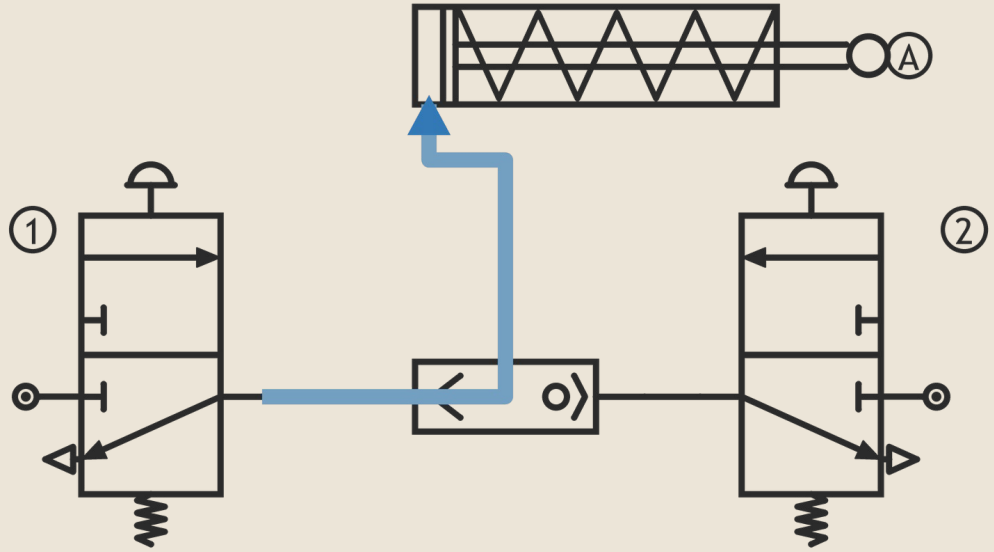


OR Control (p.19)

OR control: the piston moves when valve 1 **or** valve 2 is actuated.

The two valves are joined by a **shuttle valve**.

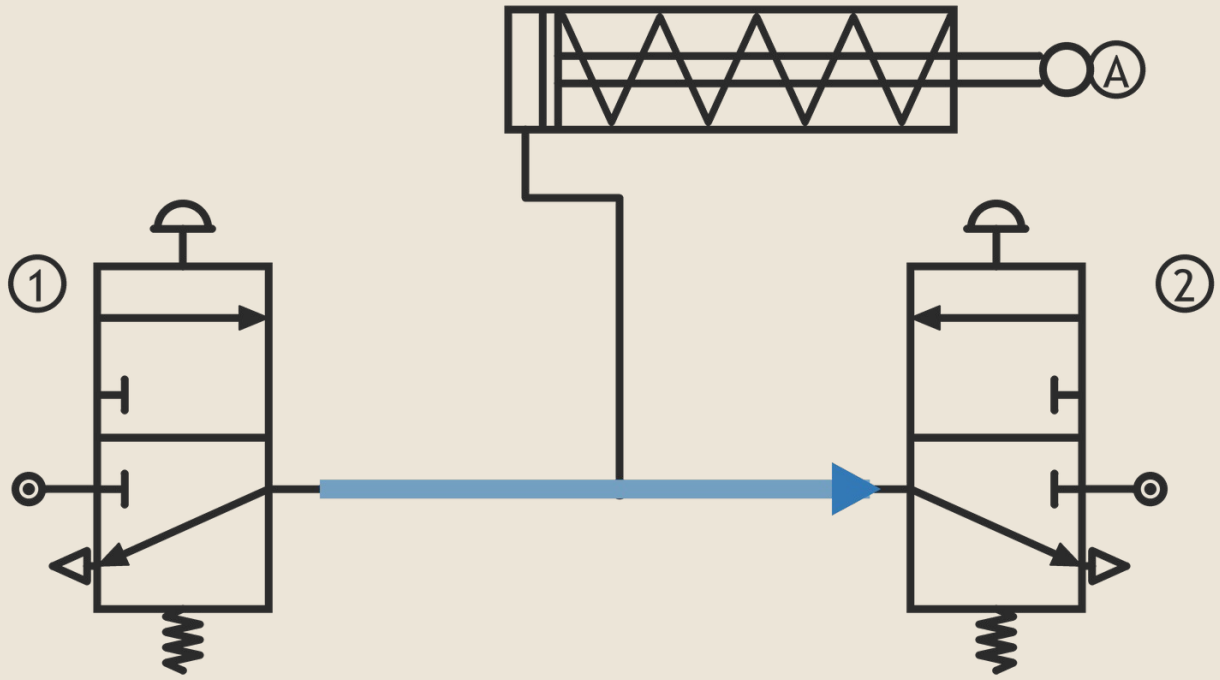
Air from either valve pushes the ball across. The air goes to the output, not out through the other valve's **exhaust**.



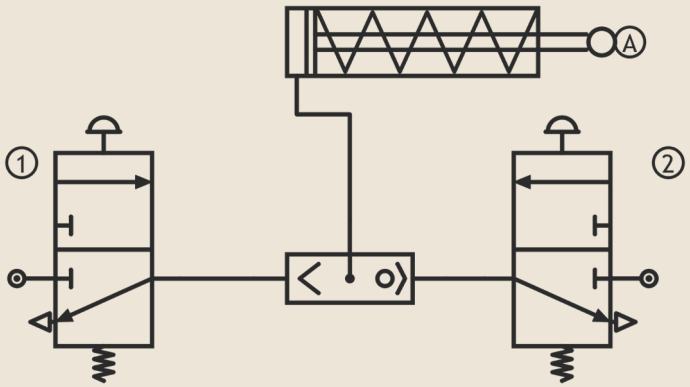
Not a T (p.19)

Do not join the two pipes with a T: that is not OR control, and it gets no mark.

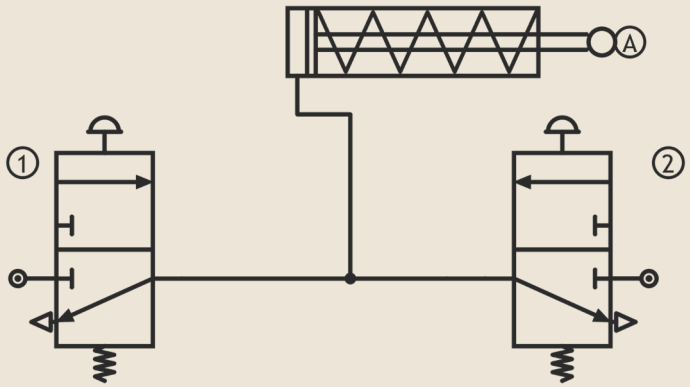
The air **escapes** through valve 2's **exhaust**.



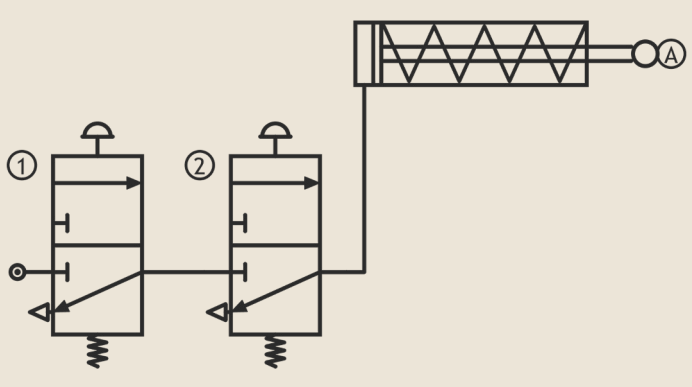
Show Me: AND, OR or Neither?



OR: a **shuttle** valve



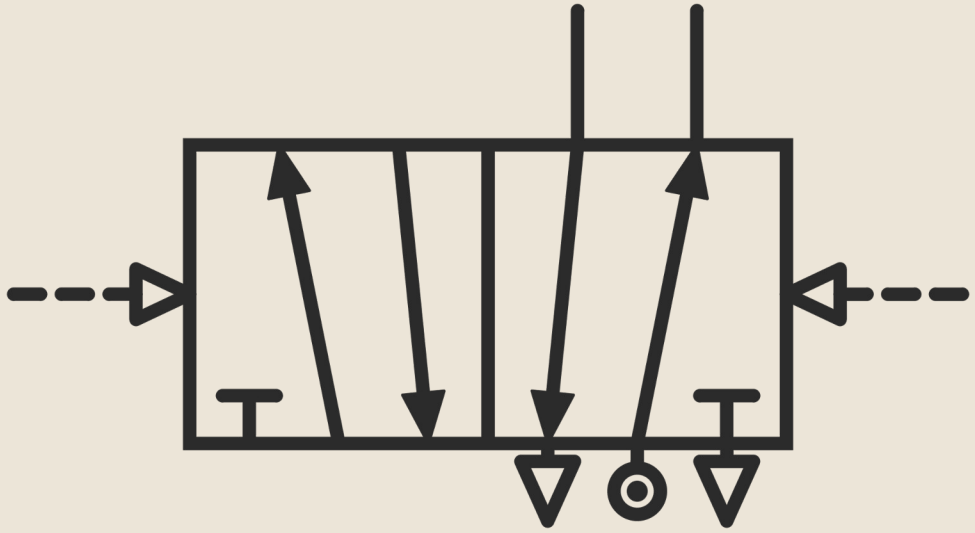
Neither: a T gets **no** mark



AND: valves in **series**

Pipe It to the 5/2 Valve (p.19)

Pipe the output of the AND valves to the **pilot** end of the 5/2 valve. Do the same with the output of a shuttle valve. Draw that pipe dashed: it is pilot air.



Words That Get the Marks (p.19)

Name it

AND control or **OR** control

OR

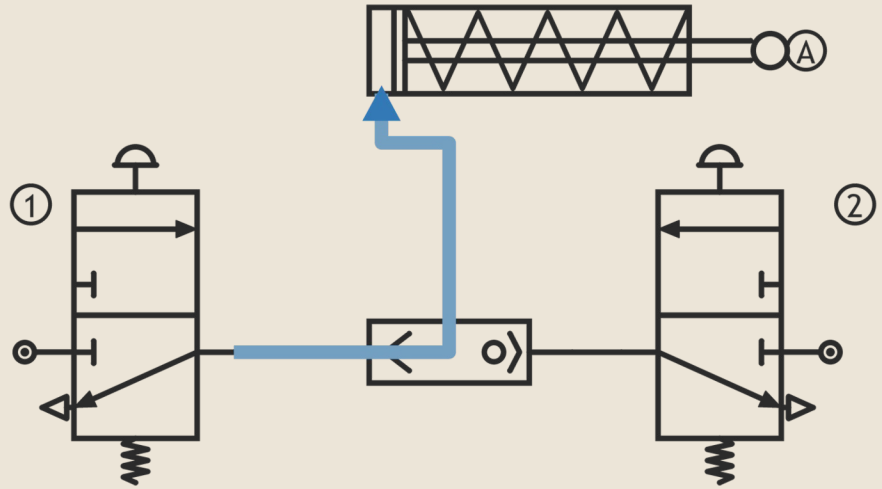
a **shuttle** valve, never a T

Pilot air

a **dashed** line into the 5/2

Every pipe

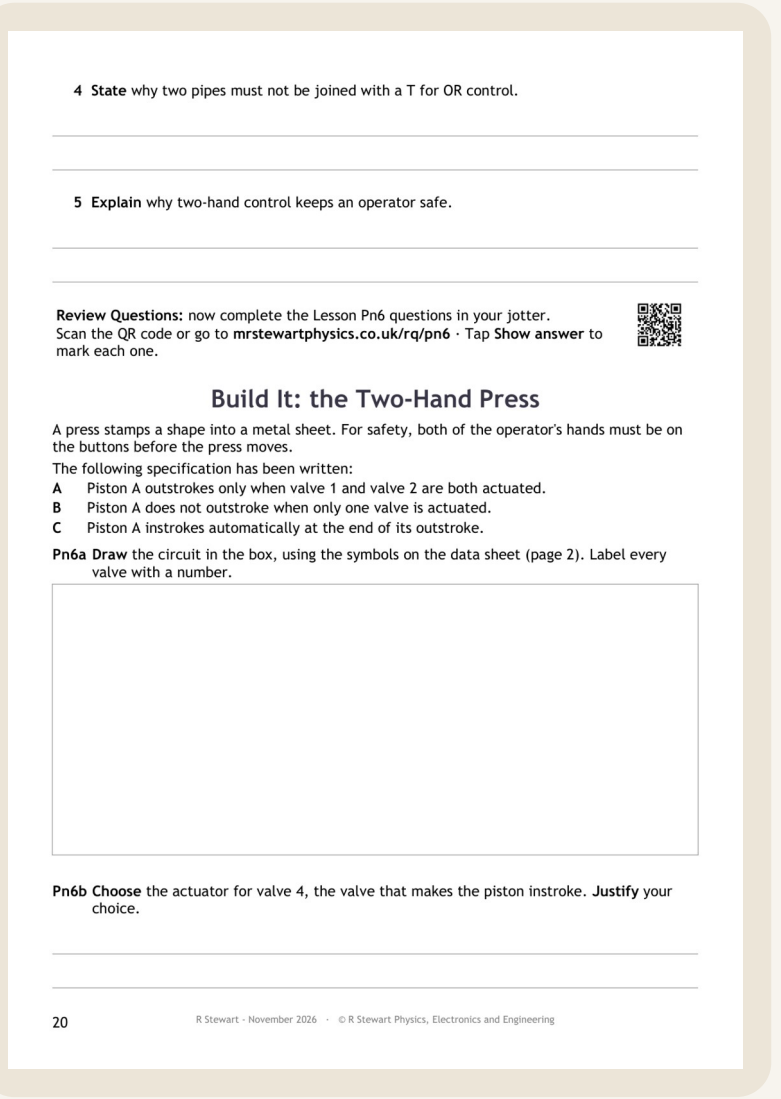
port to port



Check Your Understanding (pp 19-20)

In your booklet

Answer Questions **1** to **5** on pages **19** and **20**.



4 **State** why two pipes must not be joined with a T for OR control.

5 **Explain** why two-hand control keeps an operator safe.

Review Questions: now complete the Lesson Pn6 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn6** · Tap **Show answer** to mark each one.

Build It: the Two-Hand Press

A press stamps a shape into a metal sheet. For safety, both of the operator's hands must be on the buttons before the press moves.

The following specification has been written:

A Piston A outstrokes only when valve 1 and valve 2 are both actuated.
B Piston A does not outstroke when only one valve is actuated.
C Piston A instrokes automatically at the end of its outstroke.

Pn6a **Draw** the circuit in the box, using the symbols on the data sheet (page 2). Label every valve with a number.

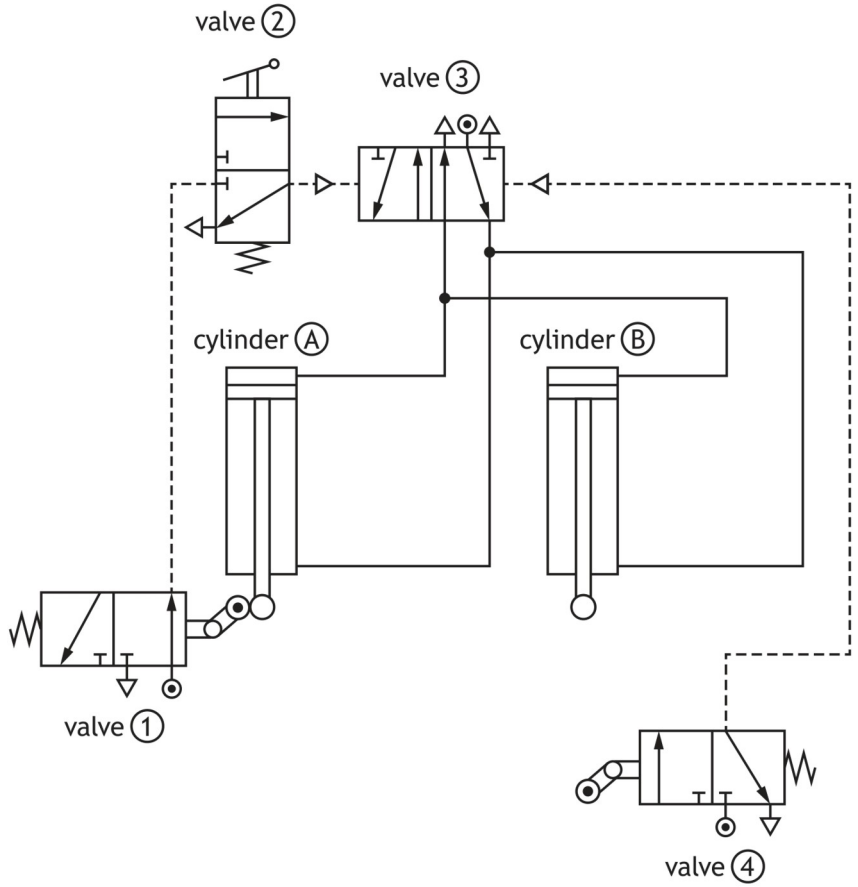
Pn6b **Choose** the actuator for valve 4, the valve that makes the piston instroke. **Justify** your choice.

Answers: Check Your Understanding (pp 19-20)

Question	Answer
1	**AND** control.
2	In **series**: valve 1 feeds valve 2's **main air**.
3	A **shuttle** valve.
4	Air escapes through the other valve's **exhaust**.
5	Hands on the **buttons**, **so away** from the piston.

Exam: 2025 Section 2 Question 8 (a)

The pneumatic circuit used in the test rig is shown below.



(a) State the type of control created by the piping of valves ① and ②. 1

Question 8 (a)

1 mark

Page 9 of the paper.

Look at how valves **1** and **2** are piped.

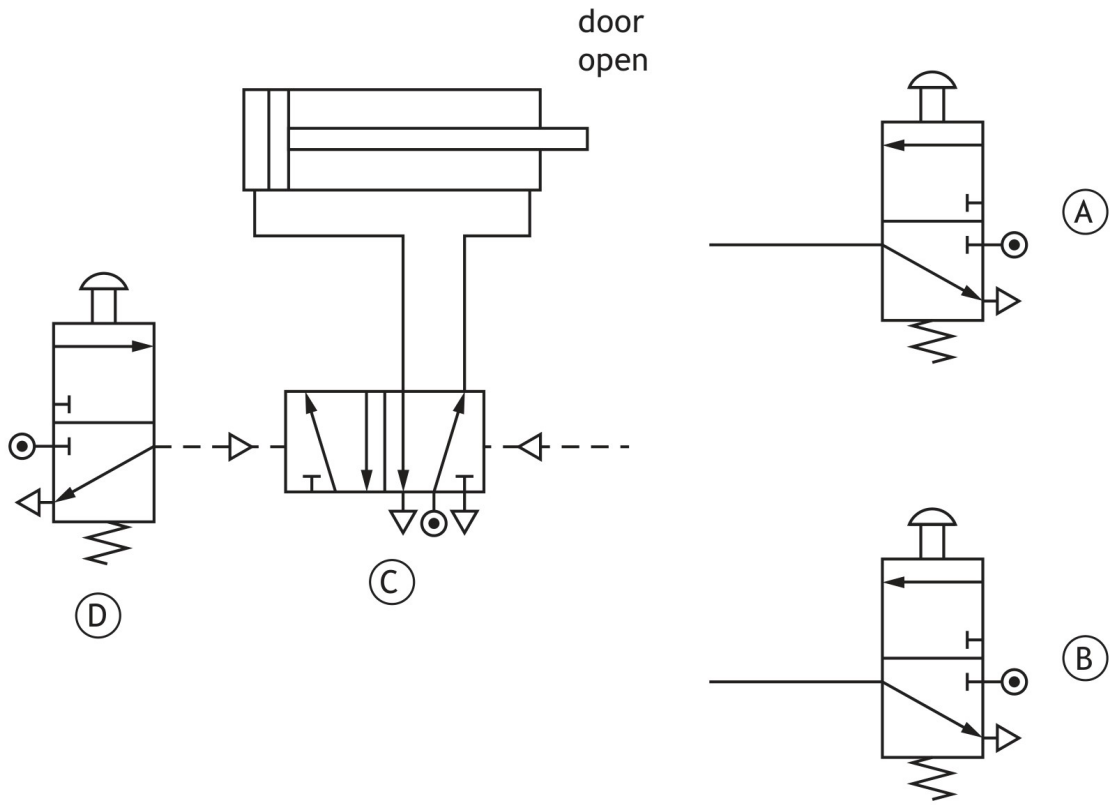
Marking: 2025 Section 2 Question 8 (a)

Question			Expected response	Max mark	Additional guidance
8.	(a)		AND (control)	1	

Exam: 2016 Section 1 Question 4 (a)

A pneumatic door in a factory is being designed.

(a) Complete the diagram below to allow the door to open when valve A **or** B is actuated. **2**



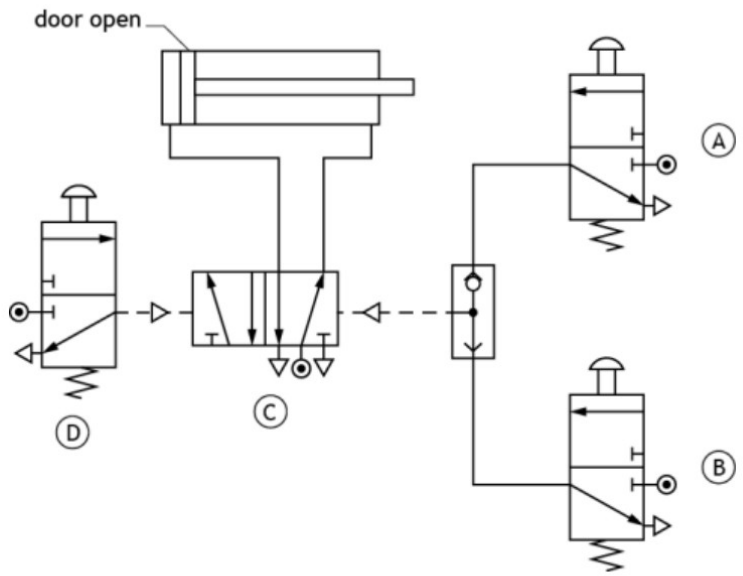
Question 4 (a)

2 marks

Page 4 of the paper.

Sketch it and **complete** it in your jotter.

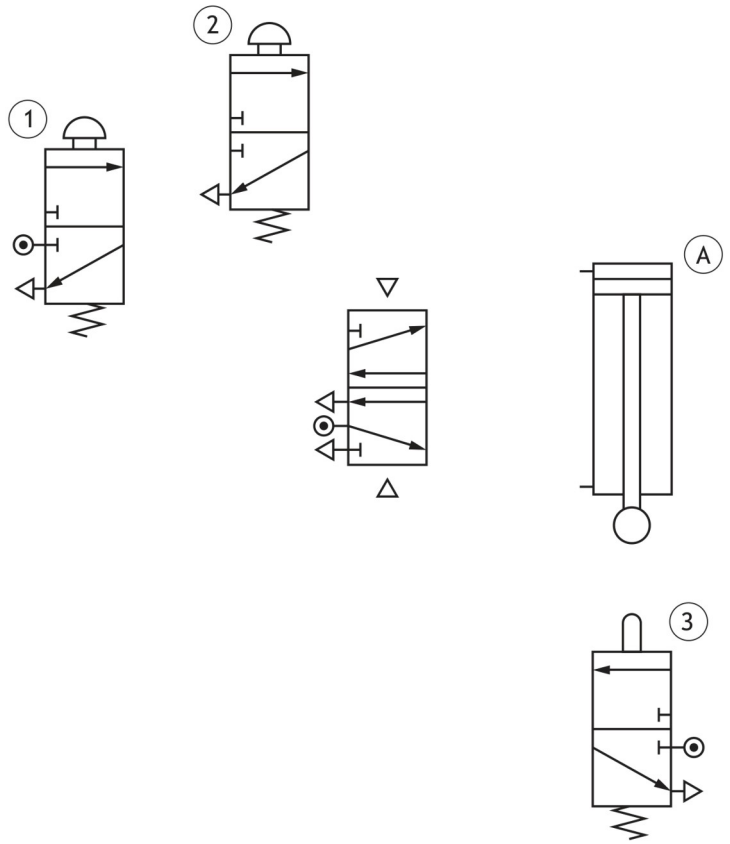
Marking: 2016 Section 1 Question 4 (a)

Question			Expected Answer(s)	Max Mark	Additional Guidance
4.	(a)			2	1 Mark for correct shuttle valve symbol. 1 Mark for all piping (port to port). If no shuttle drawn then ignore lack of T piece. Do not accept two pipes to 5/2 actuator. Ignore pilot lines.

Exam: 2022 Section 2 Question 15 (a)

A pneumatic circuit is used in part of a manufacturing process.

(a) Complete the piping of the pneumatic circuit below to outstroke the piston in cylinder (A) when valve (1) **and** valve (2) are actuated. The piston will instroke when valve (3) is actuated. **5**



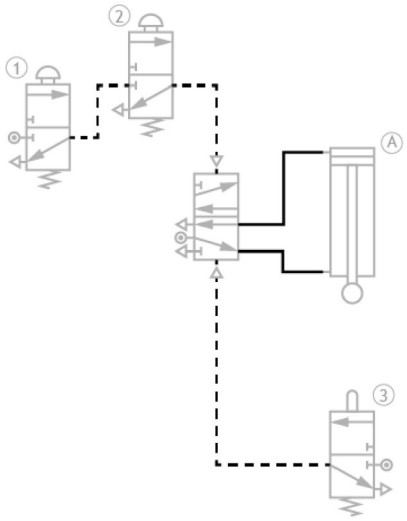
Question 15 (a)

5 marks

Page 30 of the paper.

Sketch it and **pipe** it in your jotter.

Marking: 2022 Section 2 Question 15 (a)

Question			Expected response	Max mark	Additional guidance
15.	(a)			5	Pipe connections must be port to port. 1 mark for ANDing valve ① to valve ② and piping pilot actuator on top of the 5/2 valve. 1 mark for piping up valve ③ to pilot actuator on the bottom of 5/2 valve. 1 mark for a pilot air line type for given piping into the 5/2. 1 mark for top pipe to DAC from 5/2 valve. 1 mark for bottom pipe to DAC from 5/2 valve. Allow FTE if incorrect 5/2 state outputs port are used. 1 mark max for DAC piping.

How It's Marked: Five Marks



Marks	Marking instructions
1	**ANDing** valve 1 to valve 2, piped to the **top** pilot.
1	Piping valve 3 to the **bottom** pilot of the 5/2.
1	A **pilot air line** type into the 5/2.
2	The **top** and **bottom** pipes to the cylinder. **Port to port**.

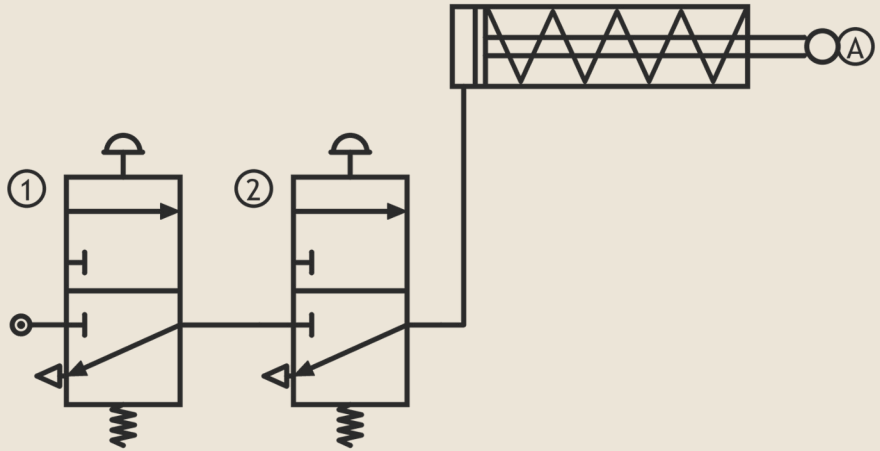
Build It: the Two-Hand Press (p.20)

A press stamps a metal sheet. Both hands must be on the buttons.

A Piston A **outstrokes** only when valve 1 **and** valve 2 are actuated.

B Piston A does **not** outstroke when only **one** valve is actuated.

C Piston A **instrokes** automatically at the end of its outstroke.



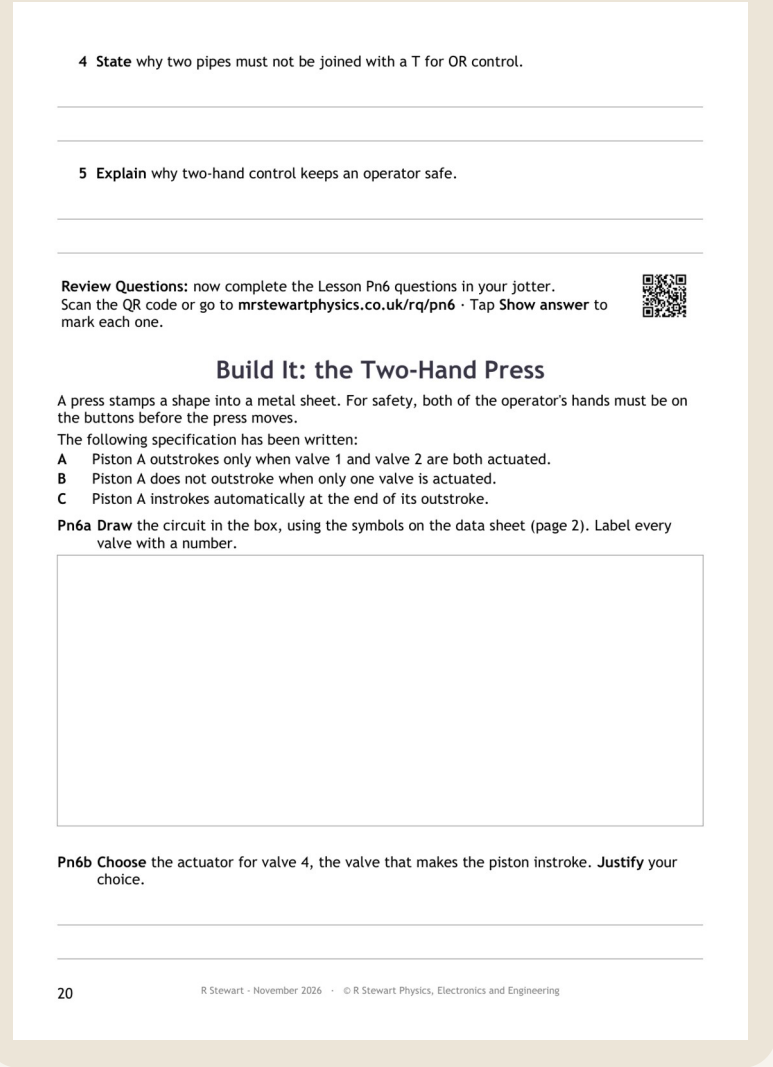
Design It (p.20)

In your booklet

Draw the circuit in the box on page **20**. Use the symbols on the data sheet (page **2**).

Label every valve with a number.

Choose the actuator for valve **4**. **Justify** it.



4 State why two pipes must not be joined with a T for OR control.

5 Explain why two-hand control keeps an operator safe.

Review Questions: now complete the Lesson Pn6 questions in your jotter.
Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn6** · Tap **Show answer** to mark each one.

Build It: the Two-Hand Press

A press stamps a shape into a metal sheet. For safety, both of the operator's hands must be on the buttons before the press moves.

The following specification has been written:

A Piston A outstrokes only when valve 1 and valve 2 are both actuated.
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C Piston A instrokes automatically at the end of its outstroke.

Pn6a Draw the circuit in the box, using the symbols on the data sheet (page 2). Label every valve with a number.

Pn6b Choose the actuator for valve 4, the valve that makes the piston instroke. **Justify** your choice.

Write a Test Plan (p.21)



Planned test	**Expected result**	**Test result**
Actuate valve 1 only, then valve 2 only.	Piston A does **not** outstroke.	
Actuate valves 1 **and** 2 together.	Piston A **outstrokes**.	
Watch piston A reach the end of its outstroke.	Piston A **instrokes**.	

Simulate It (p.21)

In your booklet

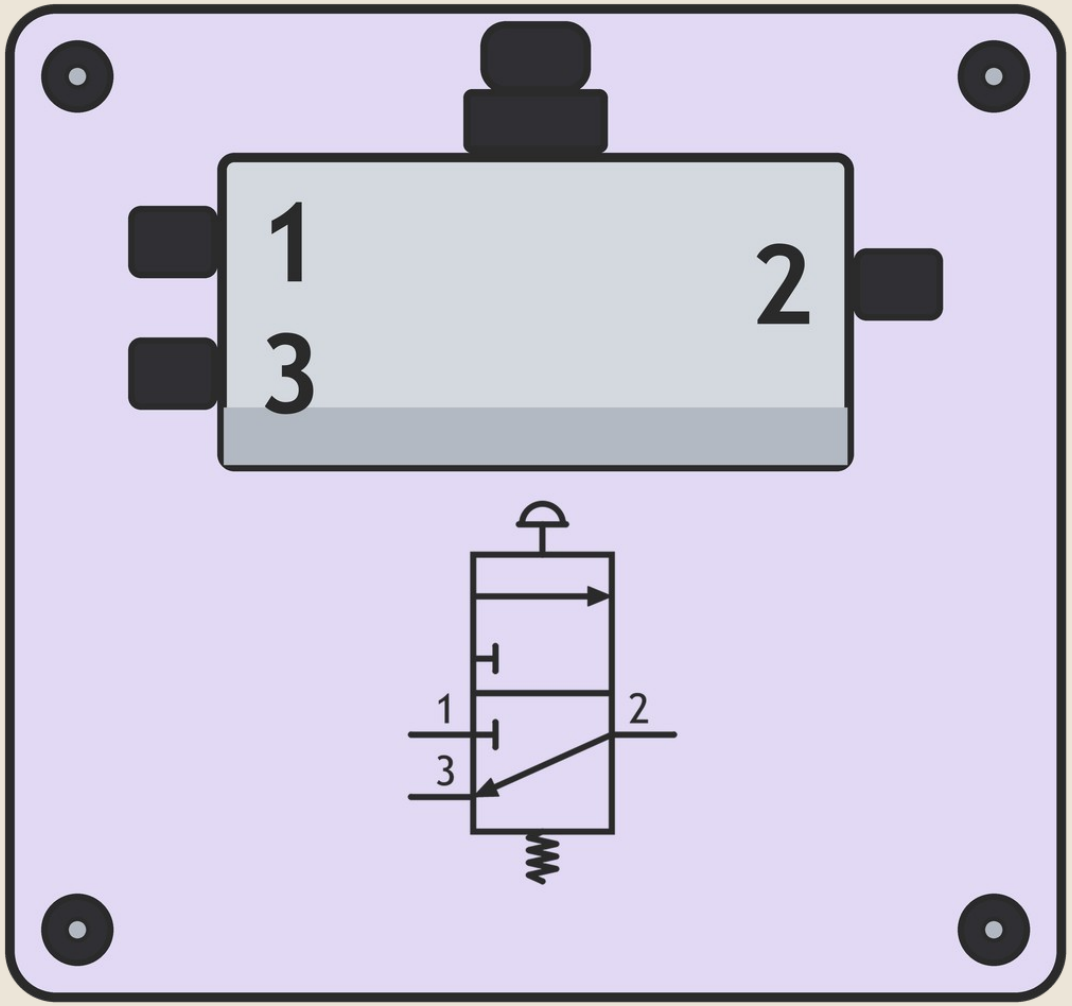
1. **Open** NoPressureSim. **Build** your circuit, as on page 8.
2. **Predict** what happens. **Test** it.
3. Air off: **build** it on the kit. Air on: **test**.

Scan me

What to Collect

Collect one set per pair.

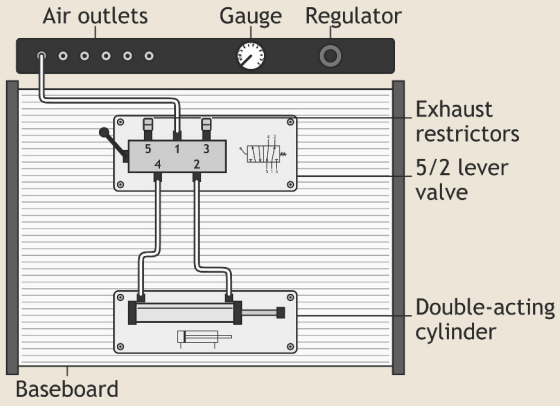
1. 3/2 push-button valve (two)
2. 5/2 pilot/pilot valve
3. 3/2 roller valve
4. Double-acting cylinder
5. Eight blue tubes
6. Goggles each



Build It on the Kit (p.21)

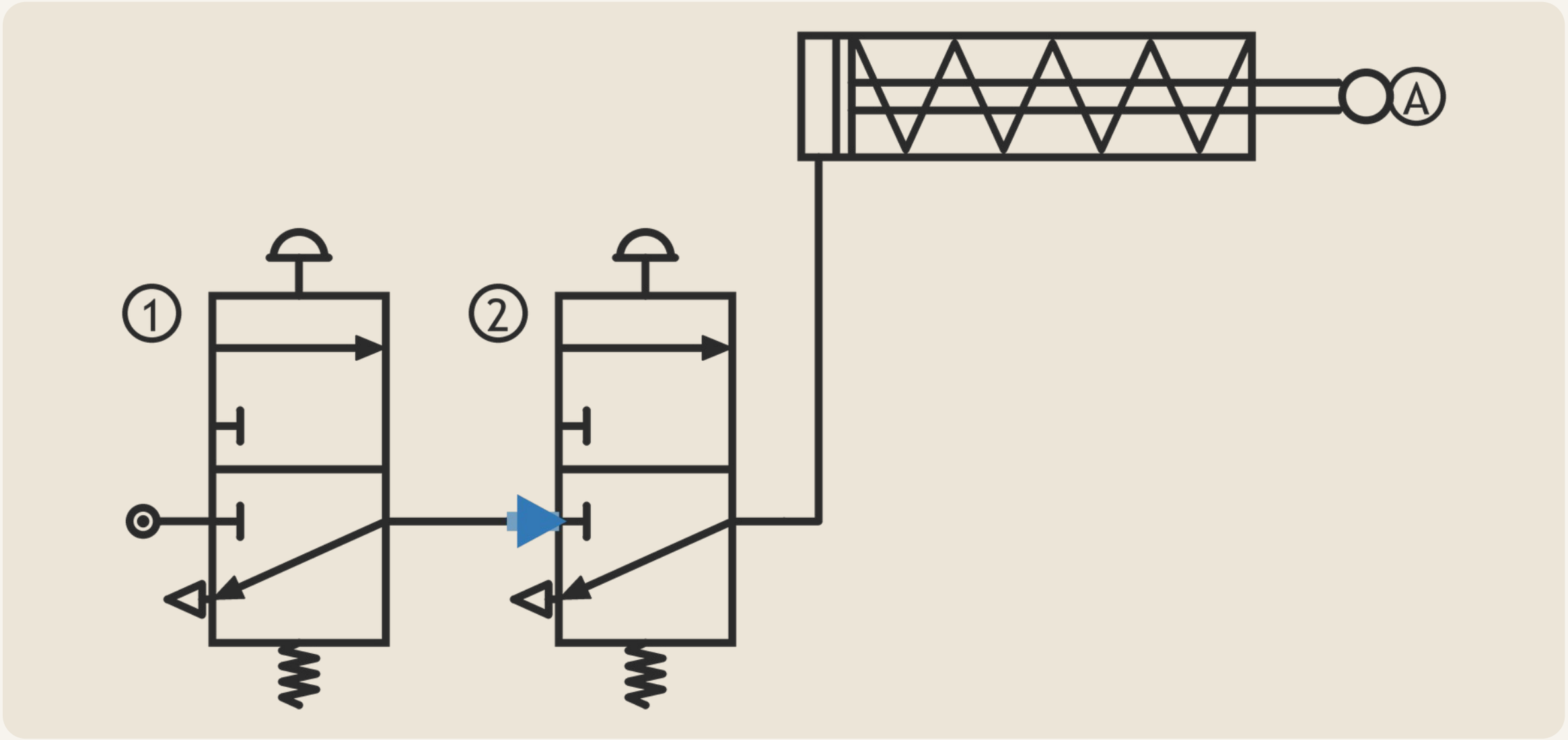
On the kit

Build your circuit, air **off**.
Valve 1's output to valve 2's **main air**.
Air on. **Test** and **record**.



SAFETY 0 bar to build. 1 to 2 bar to test.

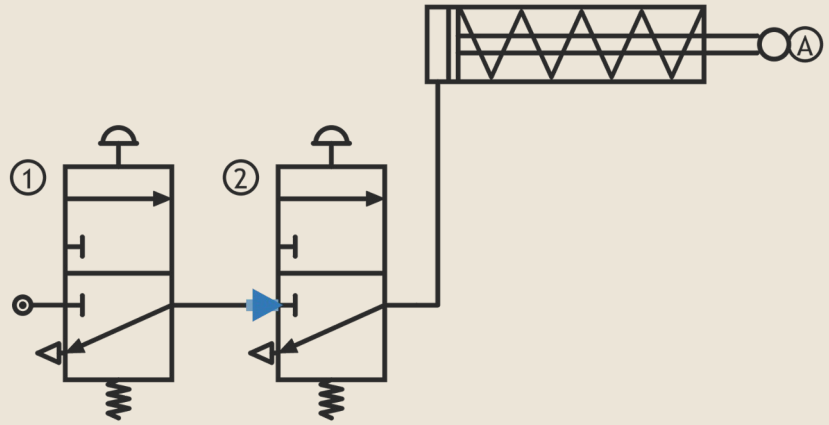
Build It: Bench Card 1



Evaluate (p.21)

In your booklet

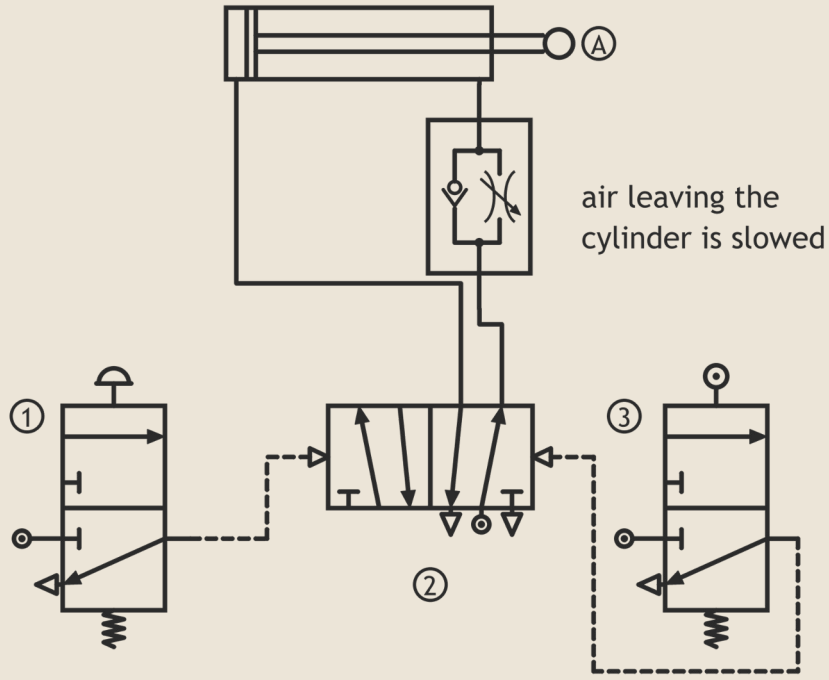
For each specification point, **state** if it was met.
Explain why, using your **test results**.
Say what you did **and** what the piston did.



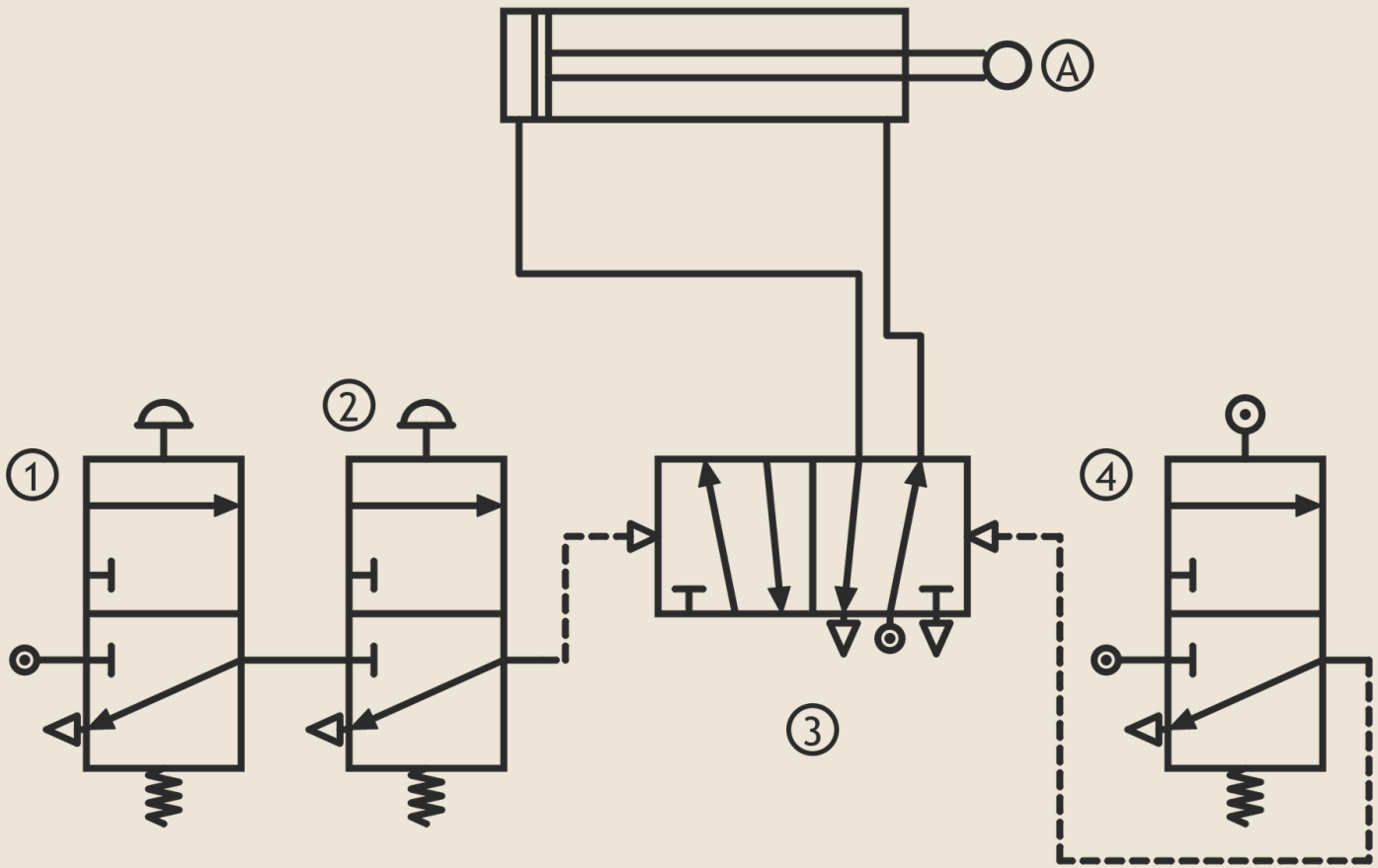
Improve and Justify (p.21)

In your booklet

The press comes down **very fast** and slams into the sheet.
Describe a modification that would solve this.
Justify it: say **why** it solves the problem.



Answers: Design It (p.20)



Valve 4

Roller: the moving piston operates it at the end of its outstroke.

Answers: Evaluate (p.21)

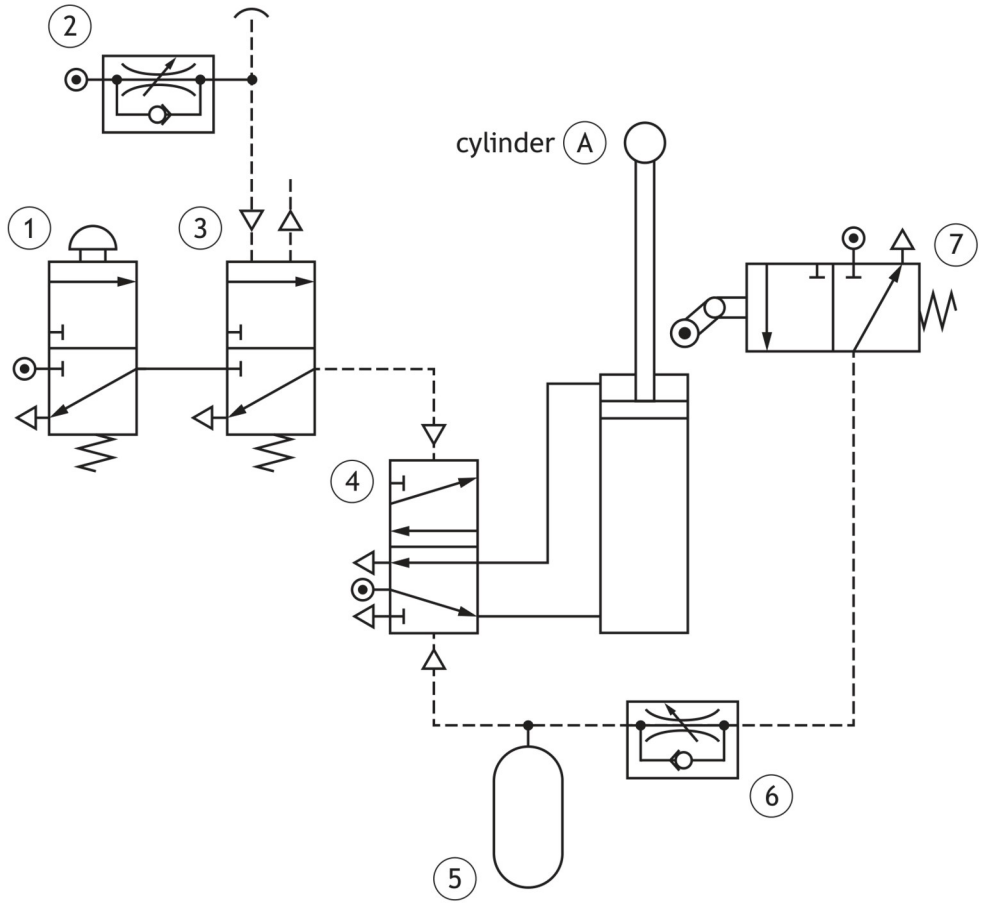
Point	Met?	Explanation
A	Yes	**In test 2**, valves 1 **and** 2 were actuated, **so** pilot air reached the 5/2 and piston A **outstroked**.
B	Yes	**In test 1**, one valve alone sent **no** pilot air, **so** piston A **did not outstroke**.
C	Yes	**In test 3**, piston A actuated valve 4, **so** the 5/2 **changed state** and piston A **instroked**.

Answers: Improve and Justify (p.21)

	Answer
Modification	A **uni-directional restrictor** slowing the **exhaust** air leaving the cylinder.
Justification	Piston A outstrokes **slowly** and **smoothly**, **so** the press does not **slam** into the sheet.

If Time: 2019 Section 2 Question 10 (b)

The pneumatic circuit used to control the movement of the saw is shown below.



Question 10 (b)

3 marks

Pages 16 and 17.

Describe the operation of the circuit.

Start: *When an increase in pressure is detected by valve 3 …*

Marking: 2019 Section 2 Question 10 (b)

Question	Expected response	Max mark	Additional guidance
(b)	*When an increase in pressure is detected by valve 3…*	**3**	
	…and when valve 1 is actuated (air is sent actuating valve 4), causing the piston to instroke…		1 mark for AND control with valve 1 and 3 causing the piston to instroke.
	valve 7 is actuated then there is a time delay…		1 mark for time delay after valve 7 being actuated.
	…then valve 4 actuates causing the piston to outstroke.		1 mark piston outstroke after valve 4 actuated and valve 7/time delay.

What Candidates Get Wrong

Name the control

most did **not** say AND (2025, 29 %)

Shuttle valve

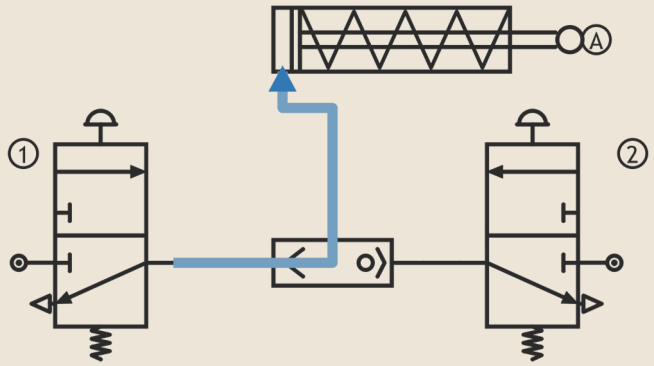
two pipes into the 5/2: **no** mark (2016 MI)

Pipe the cylinder

5/2 to **both** ends (2026)

Pilot air

a **dashed** line into the 5/2 (2017, 2022 MIs)



Review Questions (p.20)

Lesson Pn6 (online)

Answer the Lesson **Pn6** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn6**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this lesson… (p.19)



1. I can **pipe** two valves in series to give AND control.
2. I can **pipe** two valves and a shuttle valve to give OR control.
3. I can **state** the type of control a circuit uses, from its piping.
4. I can **explain** why two-hand AND control keeps an operator safe.

Scan: page 6

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 3)

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Starter: 2017 N5 Engineering Science, © Qualifications Scotland (SQA). 2017 question paper and marking instructions; 2023 N5 Engineering Science, © Qualifications Scotland (SQA). 2023 question paper and marking instructions

Exam: 2025 Section 2 Question 8 (a): © Qualifications Scotland (SQA). 2025 National 5 Engineering Science Section 2, Q8

Marking: 2025 Section 2 Question 8 (a): © Qualifications Scotland (SQA). 2025 National 5 Engineering Science Section 2, Q8 marking instructions

Exam: 2016 Section 1 Question 4 (a): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 1, Q4

Credits (2 of 3)

Marking: 2016 Section 1 Question 4 (a): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 1, Q4 marking instructions

Exam: 2022 Section 2 Question 15 (a): © Qualifications Scotland (SQA). 2022 National 5 Engineering Science Section 2, Q15

Marking: 2022 Section 2 Question 15 (a): © Qualifications Scotland (SQA). 2022 National 5 Engineering Science Section 2, Q15 marking instructions

How It's Marked: © Qualifications Scotland (SQA). 2022 National 5 Engineering Science Section 2, Q15 marking instructions

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

If Time: 2019 Section 2 Question 10 (b): © Qualifications Scotland (SQA). 2019 National 5 Engineering Science Section 2, Q10

Credits (3 of 3)

Marking: 2019 Section 2 Question 10 (b): © Qualifications Scotland (SQA). 2019 National 5 Engineering Science Section 2, Q10 marking instructions

What Candidates Get Wrong: © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2025 and 2026; 2016 S1 Q4(a), 2017 S1 Q6 and 2022 S2 Q15(a) marking instructions

Course specification: © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.6

R Stewart - November 2026